

Machine Design (EMCH327 Section 001, Spring 2022)

Mid-Term Exam 1, 8:30 AM, Friday, Feb. 11th, 2022

Name (First/Last): Jacob Vaught Score: _____/100

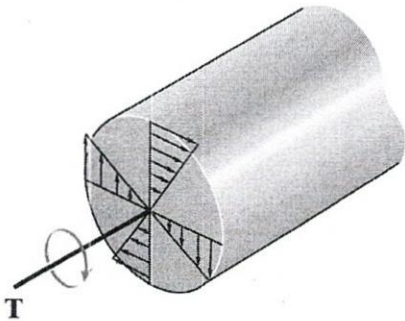
1. Single choice questions. There is one and only one correct answer for each question. ($10 \times 3' = 30'$ in total)

Q1: What is the peak of the stress-strain curve called?

A. Elasticity of modulus

☒ B. Ultimate strength

Q2: What kind of stress is induced in the cross section by a torque?



☒ A. Shear stress

B. Normal stress

Q3: What relationship applies to a composite under mechanical deformation, assuming there is no slip at the fiber-matrix interfaces?

A. The fiber and matrix have same stress, i.e., $\sigma_f = \sigma_m$

☒ B. The fiber and matrix have same strain, i.e., $\epsilon_f = \epsilon_m$

Q4: What is the ability of a material to absorb applied energy without failure?

A. Toughness

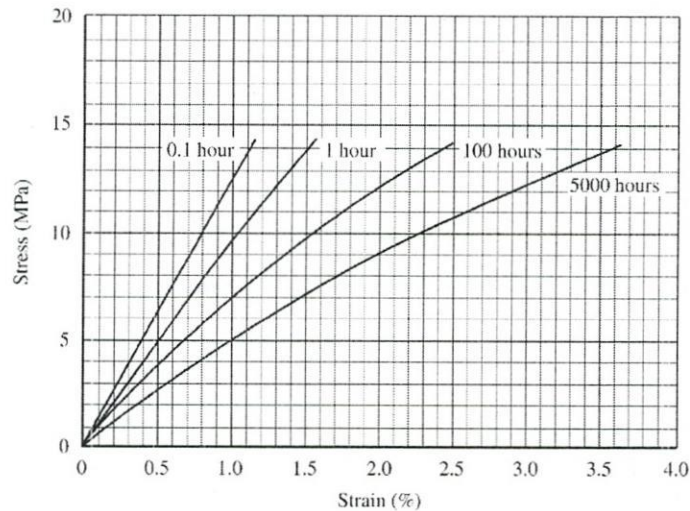
☒ B. Ultimate strength

Q5: What is the definition of specific stiffness?

A. The ratio of the tensile strength of a material to its specific weight

☒ B. The ratio of the modulus of elasticity of a material to its specific weight

Q6: What is the corresponding phenomenon for the stress-strain curves as a function of time?

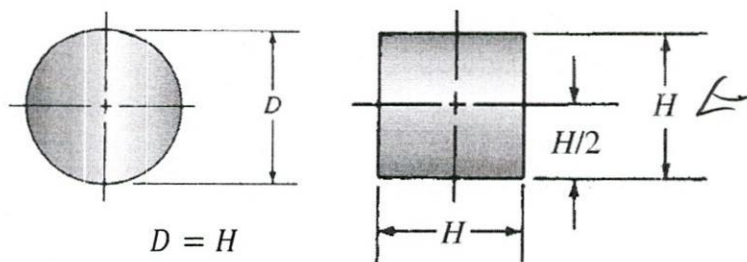


- ☒ A. Creep
☐ B. Strain hardening

Q7: In which bending test, the highest stressed part of the beam experiences no shearing force?

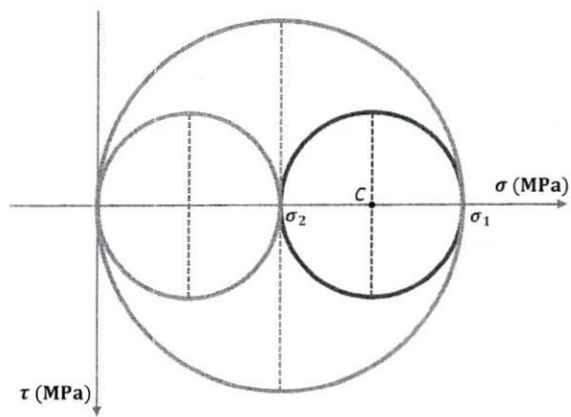
- ☐ A. 3-point bending
☒ B. 4-point bending

Q8: If the diameter of a circular cross-section is equal to the side length of a square cross-section, i.e., $D = H$, which cross-section has higher section modulus (S)?



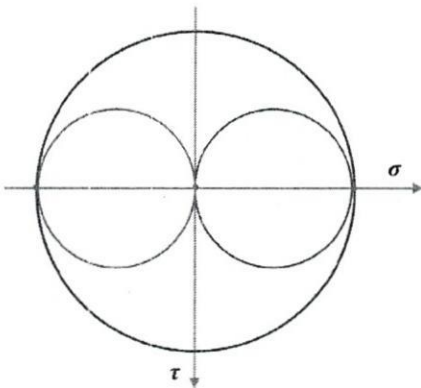
- ☐ A. Circular cross-section
☒ B. Square cross-section

Q9: For a plane stress state, if the three Mohr's circles are drawn in the figure below, then, what is the absolute maximum shear stress ($\tau_{\max}^{\text{abs}}$)?



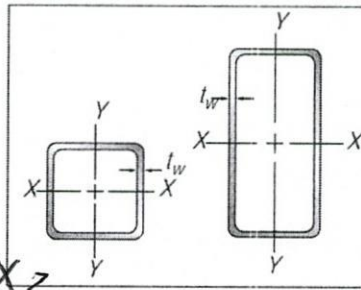
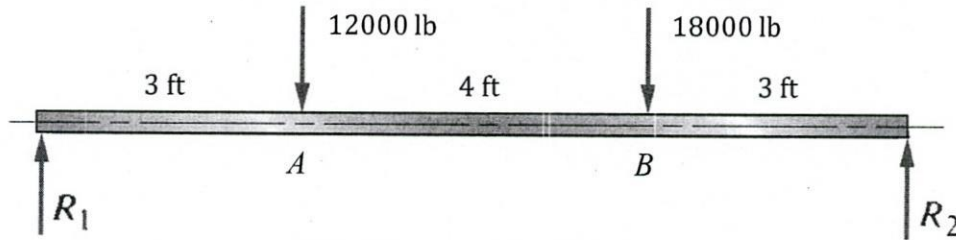
- A. $\sigma_1/2$
 B. $\frac{\sigma_1 - \sigma_2}{2}$

Q10: What kind of loading induces the plane stress that is represented by the following Mohr's circle (the two inner circles are identical)?



- A. Uniaxial tension
 B. Pure shear

2. From Table 15-14 in Appendix 15 in the textbook, select an appropriate cross-section (only need to specify the reference number in the table) for a hollow tubing to carry the load shown below with a maximum normal stress of 2730 psi. The cross-section of the hollow tubing is shown at the bottom and the bending is along the X-X axis. (30' in total)



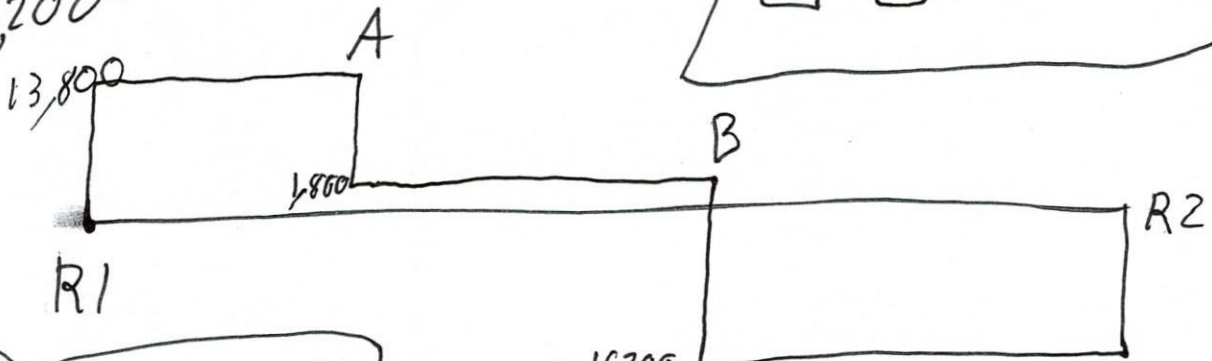
$$R_1 + R_2 = 30000 \text{ lb}$$

$$-R_1 \cdot 10 + 12000 \times 7 + 18000 \times 3 = 0$$

$$R_1 = 13800$$

$$R_2 = 16200$$

REFERENCE
[a] [c] work



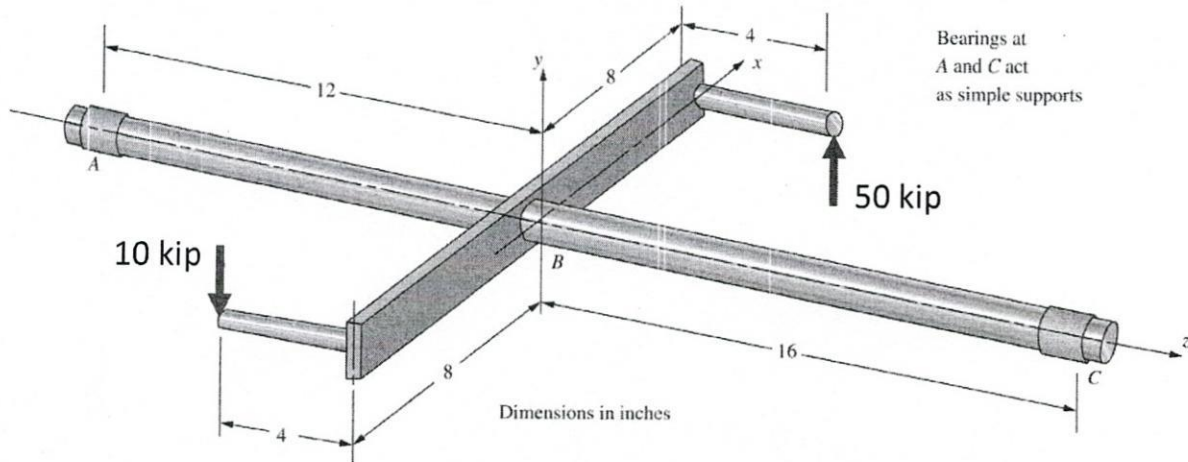
$$M_{\max} = 48,600 \text{ lb}\cdot\text{ft}$$

$$-16200$$

$$2730 \text{ psi} = \frac{M_{\max}}{S} = \frac{(48,600 \text{ lb}\cdot\text{ft}) \left(\frac{12 \text{ in}}{1 \text{ ft}} \right)}{S} \rightarrow S \approx 213.62637$$

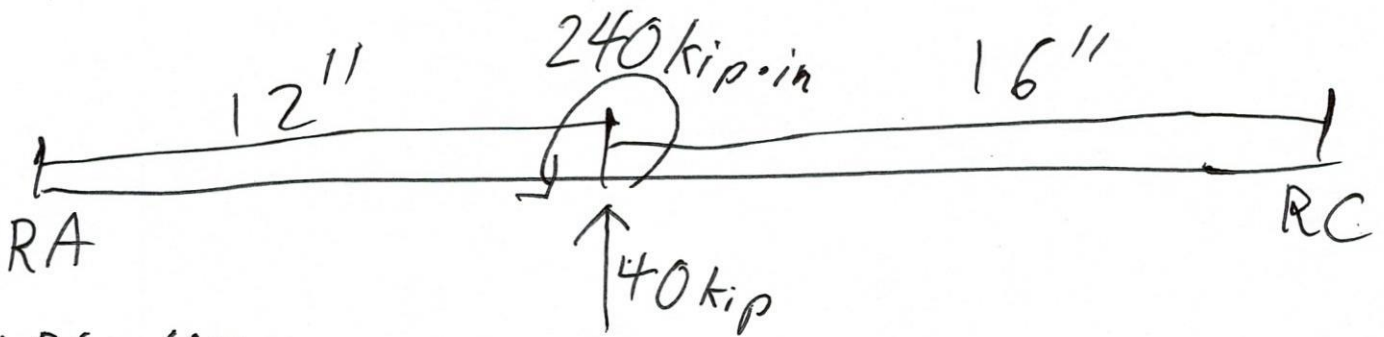
$$S \approx \boxed{17.8}$$

3. The diameter of the main shaft portion, labeled A, B, and C is 4 in. Determine the maximum tensile stress and absolute maximum shear stress for a point on the bottom of the shaft just left to the point B (in the direction to point A). Include any unbalanced torque on the shaft that tends to rotate it about the z-axis. Draw the complete shearing force and bending moment diagrams for loading in the y-z plane. (40' in total)



$$T = 60 \text{ kip} \cdot 8'' = 480 \text{ kip} \cdot \text{in}$$

$$\text{Moment at B} = 60 \text{ kip} \cdot 4 \text{ inches} = 240 \text{ kip} \cdot \text{in}$$



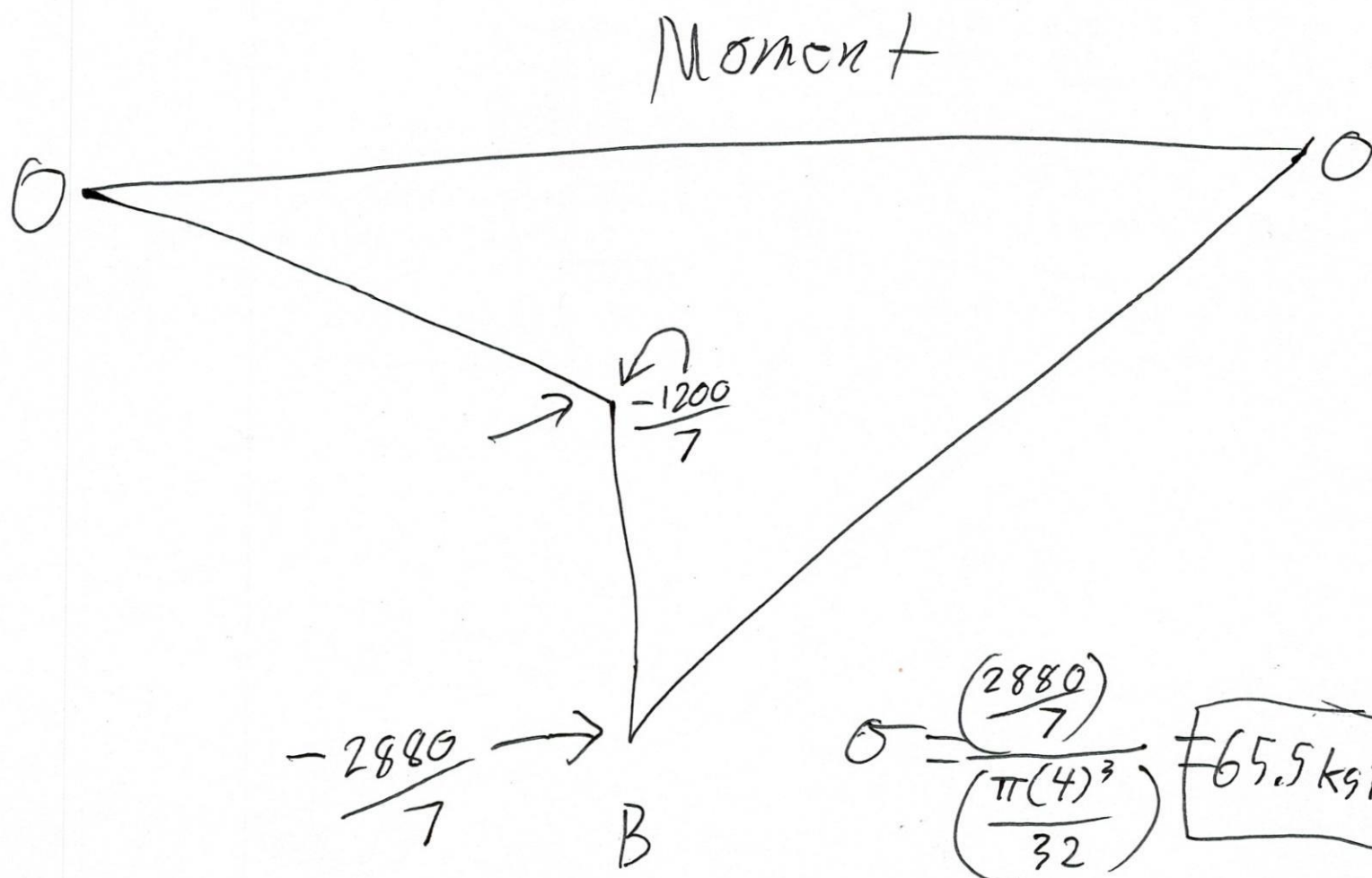
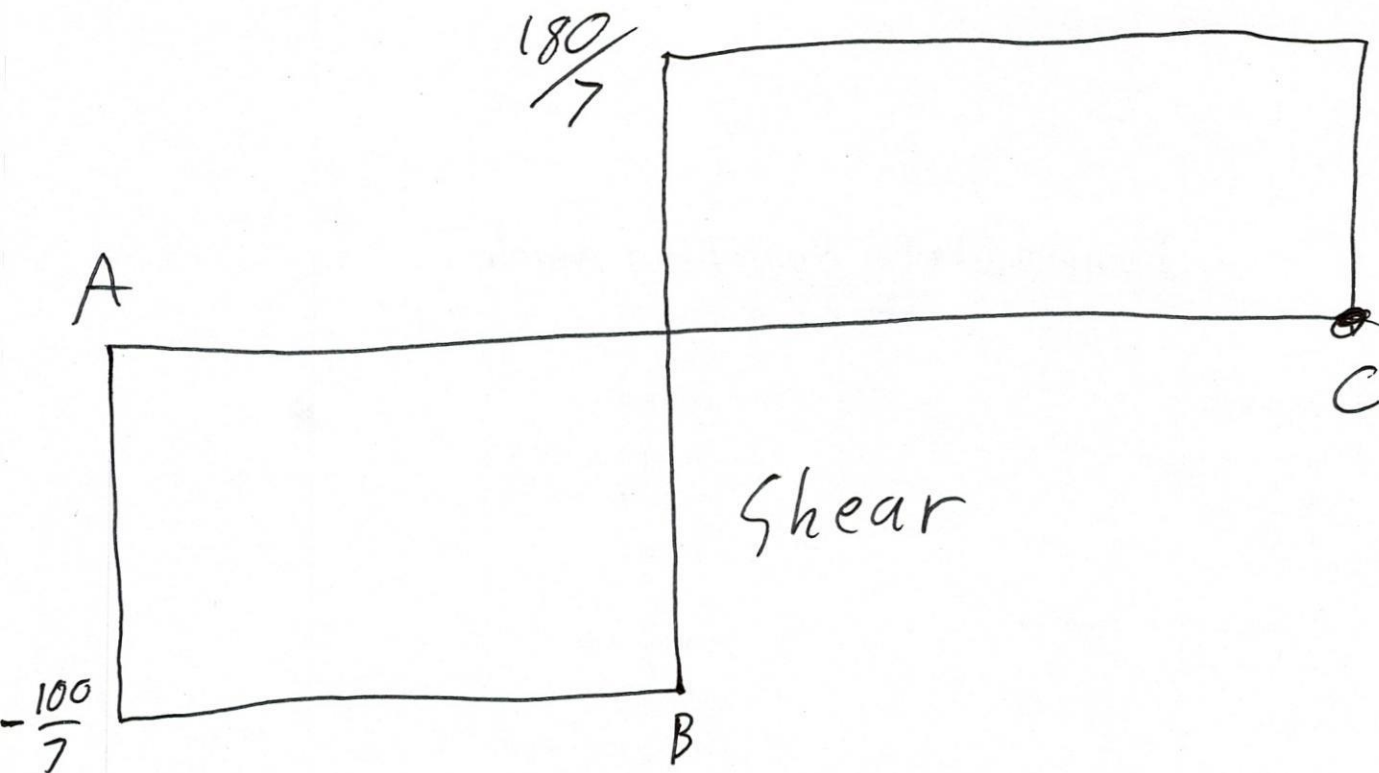
$$R_A + R_C = 40$$

$$R_A \cdot 28'' + 240 = 40 \cdot 16''$$

$$R_A = \frac{160}{7} \text{ kip}$$

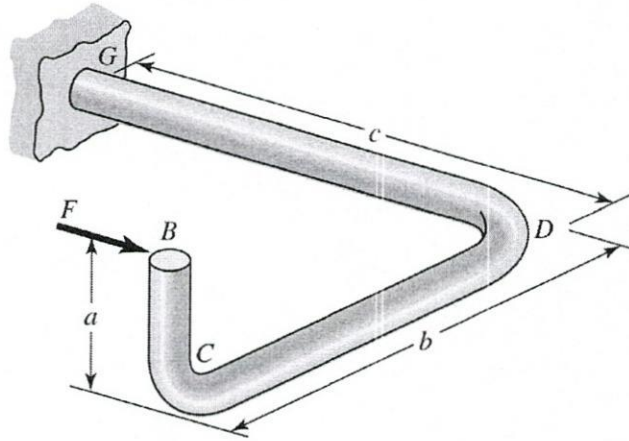
$$R_C = \frac{180}{7}$$

$$\tau = \frac{T}{Z_p} = \frac{480}{\left(\frac{\pi D^3}{16}\right)} = \boxed{38.19 \text{ ksi}}$$



$$\sigma = \frac{\left(\frac{2880}{7}\right)}{\left(\frac{\pi(4)^3}{32}\right)} = 65.5 \text{ ksi}$$

Bonus Problem Determine the total equivalent bending moment at any cross-section in the segment DG , when a concentrated force $F = 10 \text{ kN}$ is applied to the end B . Take the size $a = 0.5 \text{ m}$, $b = 2 \text{ m}$, and $c = 3 \text{ m}$. The diameter of the wire is 0.1 m . ($10'$ in total)



Draw Moment and get max between both